

APPROVED METHOD:
ELECTRICAL AND PHOTOMETRIC
MEASUREMENTS OF SINGLE-BASED
FLUORESCENT LAMPS
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-66-20(R2023)

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AN AMERICAN NATIONAL STANDARD**

Publication of this Committee
report has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by:
IES Testing Procedures Committee**



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Foreword

This approved method is a revision of *IES LM-66-14, Approved Method: Electrical and Photometric Measurements of Single-Based Compact Fluorescent Lamps*. References were updated.

1.0 Introduction and Scope

1.1 Introduction

The fluorescent lamp is an electric discharge source in which light is produced predominantly by fluorescent powders activated by ultraviolet energy generated by mercury atoms brought to an excited state by collision with electrons in a low-pressure rare-gas-mercury discharge or arc. There are two main ways to generate the electrical discharge. The most common is to pass current through the gas via electrodes (cathodes) that are connected by wires through the glass envelope. The other way is to induce a current without the need of electrodes by use of high frequency magnetic induction. This type of discharge lamp is commonly called an *electrodeless discharge lamp*.

Like most electric discharge lamps, both types of fluorescent lamps require operation in series with a current-limiting device. This device, commonly called a *ballast*, which may be either electromagnetic or electronic, limits the discharge current to the value for which the lamp type is designed. The ballast, in conjunction with the appropriate power supply, provides the required starting capability and maintains the operating lamp voltage and current as required to meet the design specifications for the lamp.

Fluorescent lamps with electrodes require long discharge lengths to be optimally efficient. Normally, the discharge lamp is a cylindrical tube with phosphor coating on the inside surface and an electrode at each end. A long tubular lamp can be made compact by being folded one or more times or spiraled in a helix in such a way that both electrodes are configured to have one connection, hence single-based construction. Further, lamps of any of these bent-tube geometries can be

covered with visible-radiation transmitting envelopes having bulb shapes such as globe (G shaped), tubular (T bulbs), reflector (R and PAR shaped) and others.

Induction-driven electrodeless fluorescent lamps, on the other hand, are necessarily more compact, as the discharge current is required to form a closed loop inside the structure. Electrodeless fluorescent lamps can be globular, A-line types, ring structures, or reflector lamps. Regardless of the electrodeless lamp's outer shape, it has a more convoluted topological construction as it has to incorporate an electromagnet outside the container for the inert gas and mercury (the gas discharge volume) around which the current circulates. Two methods are in current use: a tubular coil inserted on the axis of a globular, A-line type, or reflector lamp, or a toroidal coil through which a circular glass tube is inserted.

Finally, there are two categories of compact fluorescent lamps (CFLs): integrated and non-integrated. Non-integrated CFLs, normally only those with electrodes, are composed of a discharge tube, lamp base, and electrical connectors that interface to a lamp holder that connects the external ballast to the lamp. For non-integrated CFLs the ballast is essentially the same as for double-based (linear) fluorescent lamps, and the starting circuits can be the same, i.e., preheat, instant-start, rapid-start, or program-start. Integrated CFLs and electrodeless lamps combine the discharge tube or induction structure, ballast (normally electronic), lamp base, and appropriate housings into one structure. Because the electromagnet, which is part of the ballast, should be in close proximity to the discharge chamber, most electrodeless lamps are integrated (self-ballasted) lamps. Typically, integrated lamps can be directly connected to the branch circuit using standard lamp holders.

The electrical characteristics usually measured are line voltage, lamp current, lamp voltage, and lamp power. In the case of rapid-start lamps, the power measurements may include both the arc power and the electrode power. *Arc power* is the power consumed by the lamp discharge only and is exclusive of any power that may be supplied to the lamp electrodes from a separate voltage source. *Total lamp power* is the sum of arc power and electrode power. For this Approved Method, the determination of these data will be considered *electrical measurements*.

The photometric information usually required is total luminous flux (lumens), luminous intensity (candelas) in one or more directions, chromaticity values, and color rendering information. For this Approved Method, the determination of these data will be considered *photometric measurements*.

For special purposes, it may be desirable to determine the characteristics of lamps when they are operated at other than the standard conditions described in this Approved Method. Where this is done, such results are meaningful only for the conditions under which they were obtained. All such nonstandard conditions shall be stated in the test report. An example of a special technique for photometric testing of single-based fluorescent lamps is included in **Annex A**.

1.2 Scope

This Approved Method describes the procedures to be followed and the precautions to be observed in order to obtain uniform and reproducible measurements of the electrical and photometric characteristics of both integrated and non-integrated single-based compact fluorescent lamps, and integrated electrodeless lamps, under standard conditions in alternating current (AC) circuits (both line- and high-frequency). This document does not include linear, circular-shaped, or single-based U-bent fluorescent lamps, which are covered in ANSI/IES LM-9-20.¹ Additional information for reflector-type compact fluorescent lamps is provided in ANSI/IES LM-20-20.²

2.0 Normative References

2.1 ANSI/IES LS-1-20

Nomenclature and Definitions for Illuminating Engineering. New York: Illuminating Engineering Society; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Feb 25).

2.2 ANSI/IES LM-54-20

Approved Method: IES Guide to Lamp Seasoning. New York: Illuminating Engineering Society; 2020.

3.0 Definitions

The units of electrical measurement used in this test method are the volt, the ampere, and the watt. The units of photometric measurement are the lumen and the candela. Color is specified in terms of CIE recommended systems.³

3.1 amalgam

An alloy of mercury and other materials, used to control the mercury vapor pressure in a fluorescent lamp. The alloy absorbs or releases mercury in relationship to the amalgam temperature. Many designs of compact fluorescent lamps include amalgam technology.

3.2 cold chamber or cold spot

The location inside a fluorescent lamp that exhibits the lowest operating temperature. The location and temperature of the cold spot are critical for optimization of lamp performance.

4.0 Ambient and Physical Conditions

4.1 General

It is good laboratory practice that the storage and testing of lamps should be undertaken in a relatively clean environment.

4.2 Vibration

Lamps should not be subjected to excessive vibration or shock during testing, storage, or handling.

4.3 Temperature

Ambient (room) temperature shall be measured at a point not more than 1 m (3.3 ft) from the lamp and at the same height as the lamp. The temperature sensor shall be shielded from lamp radiation and radiation from any other source.

Under standard conditions, the ambient temperature shall be maintained at $25\text{ °C} \pm 1\text{ °C}$ ($77\text{ °F} \pm 1.8\text{ °F}$).

Fluorescent lamp technology permits lamps to be designed to perform optimally under nonstandard conditions. Lamps can be tuned to produce their maximum luminous flux at elevated temperatures for special luminaires or locations. In such cases, it is desirable to test the lamp in conditions simulating its intended operating environment. The applied environmental conditions shall be recorded. The variations allowed for standard conditions apply to the actual test conditions; e.g., if testing at 45 °C (113 °F), the allowed variation is ± 1 °C. Measurements at temperatures other than 25 °C (77 °F) are considered nonstandard and shall be so noted in the test report.

4.5 Airflow

Airflow at the surface of a lamp under test will substantially alter electrical and photometric values. No discernible airflow from sources other than the device under test is allowed. This can be tested by using a single-ply tissue paper hanging in place of the lamp.

4.6 Operating Orientation

The operating position of the lamps under test should be as specified by the manufacturer. When a position is not specified, or where more than one position is specified, the lamp should be tested in the position that will be used in the application and shall be reported in the test report. Seasoning, pre-burning, and photometric measurements shall all be done with the lamp in the same position.

5.0 Electrical Conditions

5.1 Power Supply Requirements

5.1.1 Voltage Wave Shape. The power supply shall have a sinusoidal voltage wave shape such that the total harmonic distortion does not exceed 3% of the fundamental frequency when operating a purely resistive load.

5.1.2 Voltage Regulation. The root-mean-square (RMS) voltage applied to the test lamp for the duration of the measurement interval shall be regulated to within $\pm 0.1\%$ for line voltage measurements and $\pm 1.0\%$ for high-frequency measurements.⁴

5.1.3 Power Supply Impedance. The power supply impedance for low-frequency supplies, measured at the points where the ballast and lamp are connected, shall not exceed 2% of the ballast impedance. For high-frequency power supplies, it is not possible to meet the 2% limit internally in the power supply, so electronic circuits are used to keep the output voltage at the desired level. To ensure compliance with this requirement, variable autotransformers or other voltage transforming devices, where used, shall have a kVA rating of at least five times the normal lamp wattage.

5.2 Auxiliary Equipment

5.2.1 Reference Circuit Requirements.

5.2.1.1 Non-integrated Ballast. When the electrical or photometric characteristics of single-based non-integrated fluorescent lamps are being measured for rating purposes, the lamp shall be operating in conjunction with a reference ballast at the specified source voltage.⁵

The reference ballast may be a fixed-impedance ballast with the proper impedance and power factor for the lamp to be operated, or it may be a variable ballast* with resistance and inductance that can be separately varied to provide the proper total impedance and power factor for the lamp under test. The general characteristics of the reference ballasts are described in ANSI C82.3-2016.⁶ The reference ballast values for frequency, current and impedance are given in the ANSI C78.901-2016 lamp data sheet. If readings are to be taken on a lamp type for which no standard exists, the ballast used shall comply with the general requirements of ANSI C82.1-2004(R2015)⁷ or ANSI C82.11-2017.⁸

It may be desirable to determine the characteristics of lamps when they are operated by other than reference ballasts—for example, commercial electronic ballasts. Where this is done, such results are meaningful only for the ballast and circuit with which they were obtained. Such results are not directly comparable with data taken

* Variable linear reactors are available from NWL Transformers (nwl.com) and Warner Power Systems (warnerpower.com). *Note: Provision of this information does not in any way imply an endorsement of these companies. Linear reactors suitable for use as reference ballasts may be available from other sources.*

on a reference ballast. Non-reference ballasts shall be documented in the test report.

5.2.1.2 Circuits for Line Frequency Operation of Lamps with Electrodes. The measurement circuit employed for non-integrated single-based compact fluorescent lamps is shown in **Figure 5-1**. The line frequency reference ballasts used for lamp measurements are of the series-reactor type. **Figure 5-1** shows the method of connecting into the circuit a multifunction instrument capable of measuring and displaying power, current, and voltage. The potential element of the instrument is connected on the lamp side of the current measuring element. A switch, S_1 , provides a means of measuring either lamp or input voltage of the reference circuit with the same potential element. An auxiliary voltmeter, V_1 , can be used if monitoring input and lamp voltage simultaneously is desirable. However, the use of a multichannel data acquisition device (DAQ) can eliminate the auxiliary voltmeter and can be used to monitor the input and lamp voltage simultaneously.

5.2.1.3 Preheat-Start Lamps, Non-integrated, Line Frequency. Lamps of the preheat-start type usually have an internal starting circuit. The two available contact pins are connected to the operating circuit at the points "X" and "Y" (see **Figure 5-1**). (For lamps without internal starters and having four available contact pins, the lamp shall always be be operated with the pins connected to the same points in the operating circuit of **Figure 5-2**, and

the "X" and "Y" pins should be marked). This includes not only the operation of the lamp while measurements are being taken, but all other periods of operation prior to the time of measurement; e.g., during life testing or on warm-up racks. For lamps with pins arranged in one plane, connecting the two outside pins to the operating circuit as shown in **Figure 5-2** shall be followed as a standard procedure.

For preheat-start lamps, the complete test circuit needed is a combination of **Figure 5-1** connected (at points "X" and "Y") to **Figure 5-2**. If the starting switch, S_2 , is normally shunted by a capacitor when the lamp is started, this capacitor has to be disconnected before any readings of lamp characteristics are taken. If the starter and starting capacitor are integral to the lamp, then measurements are taken with these components in the circuit.

5.2.1.4 Rapid-Start Lamps, Non-integrated, Line Frequency. For rapid-start lamps, the complete test circuit needed is a combination of **Figure 5-1** connected (at points "X" and "Y") to **Figure 5-3**. This arrangement provides separate low-voltage sources of electrode heat and allows the lamp to operate in its normal manner. The electrode voltage may be supplied from a suitable transformer, shown as T_1 in **Figure 5-3**, having known losses (or from two separate transformers). The electrode transformer windings shall be connected to the rest of the circuit so that their voltage subtracts from

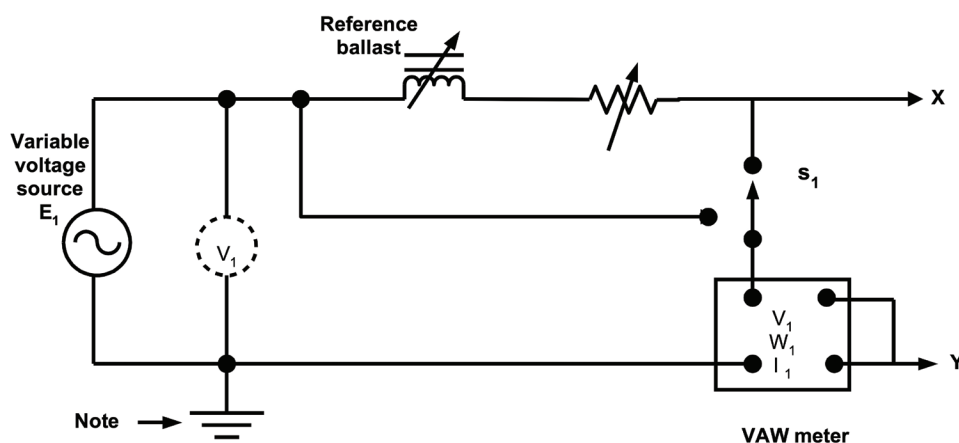


Figure 5-1. The basic measurement circuit for non-integrated single-based compact fluorescent lamps uses a variable voltage source (E_1) in series with a reference ballast. The test lamp connects to the circuit at points X and Y. Figure note: This side of the circuit shall be the one connected if a grounded supply system is used.

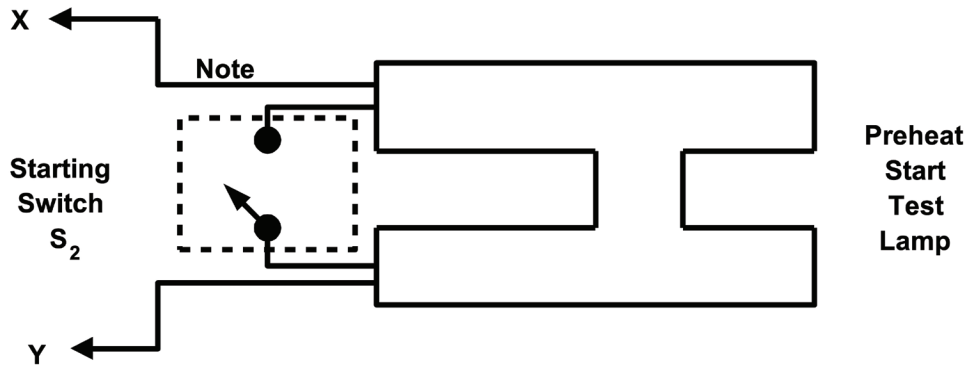


Figure 5-2. Partial circuit diagram for preheat-start lamps. Figure note: The starting circuit may be integral to the test lamp and may include a shunt capacitor.

the voltage supplied by the reference ballast circuit itself. The points "X" and "Y," therefore, will be the points of maximum potential difference at the lamp holder terminals, and the arc will operate between these two points.

The power consumed by the electrodes in an operating rapid-start lamp is the amount indicated by wattmeter W_2 , less the transformer loss. The power consumed by the discharge in an operating rapid-start lamp is the amount indicated by wattmeter and is usually referred to as *arc watts*. The total lamp power is the sum of W_1 and W_2 less the transformer loss:

$$(Total\ rapid\text{-}start\ lamp\ power) = (Arc\ power) + (Electrode\ power) - (Transformer\ power\ loss)$$

The electrode transformer in **Figure 5-3** (represented by T_1) shall be calibrated to determine its power loss under normal load conditions according to the pertinent ANSI standard.⁹

It should be noted that **Figure 5-3** shows only the connections used for operating lamp electrodes, not the connections where "dummy load" resistors are temporarily placed across T_1 's secondary windings for calibration purposes.

5.2.1.5 Instant-Start Lamps, Non-integrated, Line Frequency. For instant-start circuits, the two wires exiting each tube are shorted inside the base of the lamp. For instant-start circuits, the complete test circuit needed is a combination of **Figure 5-1** connected (at points "X" and "Y") to **Figure 5-4**. The starting circuit for

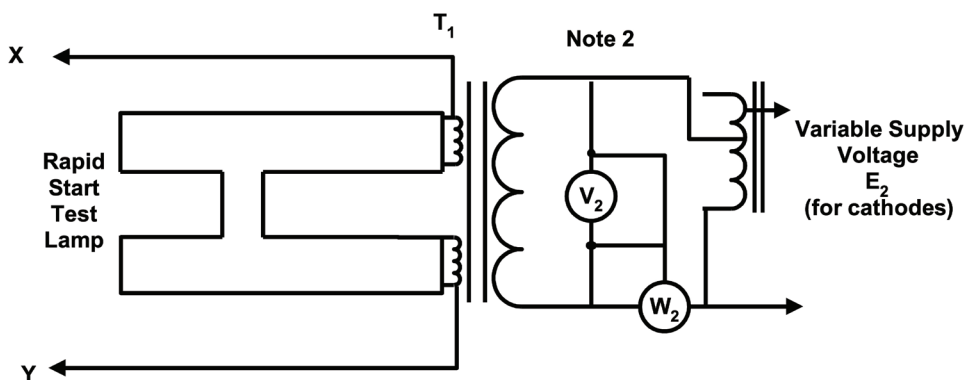


Figure 5-3. Partial circuit diagram for rapid-start lamps. The complete test circuit is a combination of Figure 5-1 and Figure 5-3. Figure Note 1: The source voltage E_2 shall come from the same line (and the same phase if a three-phase supply is being used) as the reference ballast supply voltage E_1 . Unless this precaution is observed, the subtractive polarity of the electrode windings may not actually be obtained.

instant-start lamps requires higher voltage than preheat or rapid-start lamps, as no benefit of electron emission from the electrodes is available.

5.2.2 Circuits for High Frequency Operation of Lamps with Electrodes. Lamps intended for operation on high frequency are normally four-pin lamps without an internal starting circuit. Starting is accomplished by the electronic ballast circuit, which can be preheat, instant-, or program-start. High-frequency reference ballasts are of a non-inductive resistor type. A variable power source capable of providing the voltage and frequency specified for the reference circuit is required. The lamps shall always be operated with the pins connected to the same points in the operating circuit, as described in **Section 5.2.1.4**. Wires shall be of low-loss, low-capacitance types, and shall be kept as short as possible to minimize the effect of the parasitic capacitance. Components, such as switches or relays, shall not be permitted on the lamp side of the high-frequency reference ballast.

5.2.3 Integrated Compact Fluorescent Lamp and Electrodeless Fluorescent Lamps. The measurement circuit employed for integrated compact fluorescent lamps, including integrated electrodeless fluorescent lamps, is shown in **Figure 5-5**. Usually, a variable power source capable of providing the specified AC voltage is needed to provide the input voltage to the lamp. This may be an electronic power supply, or a separate step-up transformer operated from regulated line voltage.

Figure 5-5 shows the method of connecting into the circuit a multifunction instrument capable of measuring and displaying wattage, current, and voltage. The potential element of the instrument is connected on the lamp side of the current measuring element.

Integrated fluorescent lamps generally use a medium screw-base to allow the lamp to replace an incandescent source. For photometry of a screw-based lamp, a Kelvin (four-point) lamp holder shall be used. A Kelvin lamp holder has two input power leads providing the voltage

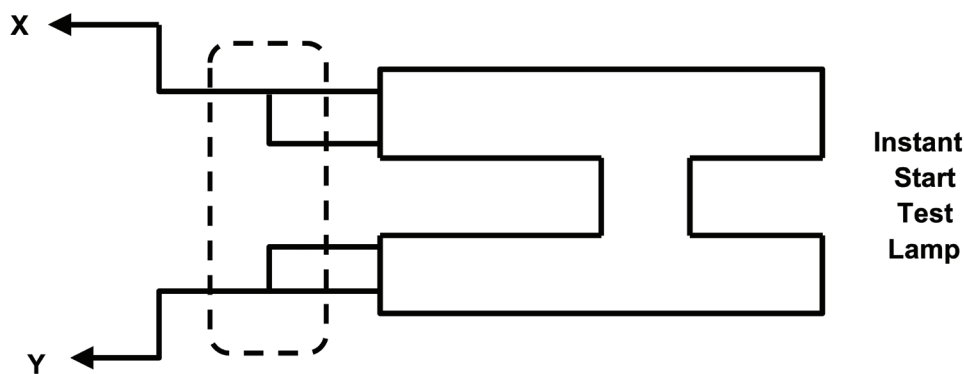


Figure 5-4. Partial circuit diagram for instant-start lamps.

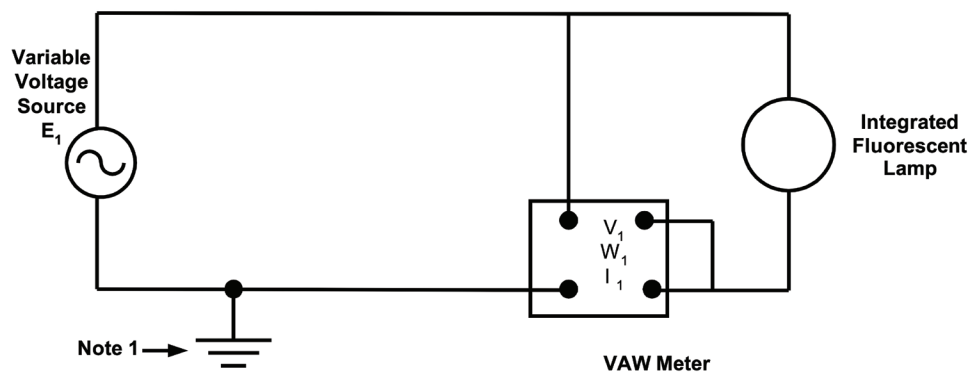


Figure 5-5. Test circuit for a single-based integrated fluorescent lamp. *Figure note 1:* This side of the circuit shall be grounded if a grounded supply system is used.

to the lamp base, and two sense leads for measuring the voltage at the lamp base.

5.2.4 Starters. If required, starters shall comply with the specifications in ANSI C78.180-2003(R2016).¹⁰

5.3 Electrical Instrumentation

5.3.1 Frequency Response. Distorted waveforms are present in discharge lamp circuits; therefore, all test instruments shall respond adequately to frequencies higher than the input line fundamental. The instrument(s) chosen for voltage and current measurements shall be of the true-RMS type and shall have a specified tolerance and frequency response adequate to ensure that these requirements are met. For high-frequency measurements, instruments shall have minimum frequency response of 100 kHz.

5.3.2 Impedance Limitations. Voltage inputs of the multifunction meter shall have input impedances greater than 1 megaohm (MΩ), and the current inputs shall have impedances less than 20 mΩ.

5.3.3 Instrument Tolerance. Instrument manuals provide values for the tolerance (accuracy) of the instrument, normally as a percentage of a reading. The instrument tolerance shall be $\pm 0.5\%$ or less for voltage and current and $\pm 0.75\%$ or less for the wattage. The reader is referred to the GUM and NIST 1297 for conversion of instrument tolerance to the measurement uncertainty.^{11,12}

6.0 Testing Procedures Requirements

6.1 Preparation for Test

6.1.1 Marking and Handling Requirements. Marking and handling requirements are found in *ANSI/IES LM-54-20, Approved Method: IES Guide to Lamp Seasoning* (refer to **Section 2**).

6.1.2 Seasoning. Prior to taking initial measurements, all new single-based fluorescent lamps selected for testing shall be seasoned per *ANSI/IES LM-54-20, IES Approved Method: IES Guide to Lamp Seasoning*.

6.1.3 Pre-burning. In order to limit the time for the subsequent stabilization in the measurement system, pre-burning should be performed. Non-integrated compact fluorescent lamps should be operated on an appropriate commercial ballast at the rated ballast voltage $\pm 5\%$; integrated compact fluorescent lamps, including integrated electrodeless lamps, should be pre-burned at rated voltage $\pm 5\%$. Typical pre-burning time is 5 to 16 hours. The standard orientation is base up unless otherwise specified. Pre-burning shall be performed at a temperature that is within $\pm 5^\circ\text{C}$ of the ambient temperature at which testing will be performed.

The pre-burning requirement may be met during the seasoning period if mercury in the lamp is not disturbed prior to taking measurements.

6.1.4 Transfer of Lamp to the Measurement Circuit. The test lamp is extinguished at the end of the pre-burning period and then transferred to the test position in the photometric measurement system. The handling of a lamp during transfer can cause a change in the location of the condensed mercury. The test lamp shall be transferred such that its orientation remains the same as during pre-burn at all times, and the lamp shall not be subjected to shock or vibration, so as to minimize the change in lamp operating characteristics between pre-burn and measurement circuits.

6.2 Stabilization

6.2.1 Preferred Method. Once pre-burning is complete and the lamps have been transferred to the test position in the photometric measurement system, the lamps shall be operated so as to attain stable light output and electrical usage. The time required to attain stabilization can vary considerably, depending on whether the lamps are transferred hot or cold from the pre-burn position, the lamp wattage, whether the lamps are covered or bare, and the lamp design, i.e. amalgam or non-amalgam. Typical stabilization times for a range of lamp

designs and with normal care in handling of the lamps are 5 to 45 minutes.

Variation in the acceptance criteria and the number and duration of the measurement interval provide different levels of lamp stability, either more or less severe. Should a different method or criterion be used, it shall be so noted in the test report.

The following four-step method for determining whether a fluorescent lamp is stable shall be used:

Step 1. Take six measurements of the lamp light output at one-minute intervals (total time = five minutes).

Step 2. Determine the maximum measured value, the minimum measured value, and the average value of the six consecutive measurements. Calculate the percent stability by dividing the difference between the maximum and minimum values by the average value of the six consecutive measurements.

Step 3. If the value calculated in *Step 2* does not exceed 1.0 %, the lamp is considered stable.

Step 4. If the value calculated in *Step 2* exceeds 1.0 %, the lamp is considered unstable, and measurements in 1-minute intervals have to be continued until the last six measurements meet the criterion in *Step 3*.

6.2.2 Unusual Conditions. A lamp that shows visual non-uniformities, e.g., striations, swirling or other abnormal behavior, is considered unstable. Swirling can usually be detected with the naked eye; however, there are also instabilities that occur in the lamp that are not visible and that may result in abnormal photometric and electrical measurements. If such instabilities are suspected to be present, running a small permanent magnet along the length of the lamp may alleviate the instability. Unstable lamp voltage may also be due to such arc non-uniformities even when not visually observed. Turning the lamp off for approximately 15 seconds without disturbing the lamp position and then restarting it will usually eliminate lamp striations or swirling. The stabilizing procedures should then be resumed before any measurements are made.

6.3 Photometric Measurement Requirements

6.3.1 Integrating Sphere Measurement. The integrating sphere method gives total luminous flux of the test lamp with one measurement. Air movement is minimized, and the temperature within the sphere is not subject to the fluctuations usually present in a temperature-controlled room. However, accumulated heat from the lamp under test may raise the temperature within a small sphere. To maintain an ambient temperature of $25\text{ }^{\circ}\text{C} \pm 1\text{ }^{\circ}\text{C}$ ($77\text{ }^{\circ}\text{F} \pm 1.8\text{ }^{\circ}\text{F}$), it may be necessary to keep the sphere open while the lamp is stabilizing, and then close it just prior to taking the measurement.

The reader is referred to ANSI/IES LM-78-20 for potential sources of error and methods for correction.¹³

6.3.2 Intensity (Candela) Distribution. The distribution of luminous intensity of a lamp may be determined with a goniophotometer. Since most single-based fluorescent lamps require that all excess mercury be condensed in pockets or cooling chambers for proper stabilization, it is critical that these lamps remain in a fixed orientation with respect to gravity. Goniophotometers shall be the type that maintains the burning position unchanged with respect to the local gravitational vector, while motions in other directions are acceptable. Only Type C goniometers are allowed. The readings are generally taken in five-degree, or finer, vertical increments and 22.5-degree, or finer, lateral increments, in order to avoid the inaccuracies that can result from rapidly changing or irregular-form distributions.

Potential sources of errors that require consideration are:

- Stray light
- Distance measurement
- Deviations from the inverse-square law (with respect to angular size of source)
- Angular alignment of lamp and detector
- Errors in the scan angles
- Spectral errors due to spectral reflectivity of turning mirror(s)
- Draft induced by the goniometer movement

6.4 Color Measurements

The measurement of color is specified in terms of chromaticity coordinates³ and color rendering indices^{14,15} as defined by CIE.

7.0 Test Report

The test report shall list all significant data for each lamp tested, together with performance data. The report shall also list all pertinent data concerning conditions of testing, type of equipment, and types of lamps being tested. Typical items reported may include:

- Date and testing agency
- Manufacturer's name and lamp designation
- Type of test
- Number of lamps tested
- Description of lamps tested
- Rated electrical values for the lamp tested
- Number of hours the lamps operated and operating cycle
- Ambient temperature
- Lamp position during test
- Pre-burning and stabilization times
- Type of standard lamp used and correction factors, if any
- Photometric method
- Sphere diameter or test distance
- Circuit conditions and fixed parameters
- Measured light output (lumens and/or candelas) and electrical values (volts, amperes, watts) of each lamp, and group averages
- Color data
- Spectral power distribution
- Test equipment used
- Statement of uncertainty
- Any deviations from standard operation procedures
- Observations of any unusual behavior of the lamps (e.g., range of measured lumen values when the stability reference limit is exceeded)
- Any deviations from the criteria recommended in this Approved Method

Annex A – Photometric Measurements of Lamps Using Fixed-Current Method at High Frequency

Note: The information in this Annex is not intended to be a specific recommended procedure but is presented as reference information to more fully acquaint the user with some alternative methods of photometric measurements of non-integrated compact fluorescent lamps. Because of the wide range of potential applications for these lamps, there exists a strong interest in expanding or modifying the measurement procedure to obtain additional information on electrical and photometric characteristics.

Commercial electronic ballasts for use with non-integrated compact fluorescent lamps are usually designed to keep the lamp current at the nominal level specified in the relevant lamp data sheet. Therefore, it may be necessary to perform photometric measurements that are made at other than the nominal current.

For fixed-current measurements, the setup is similar to that used for the reference measurement, but the current to the lamp is controlled manually by an electronic control circuit, or by control software. Use of the reference ballast specified in the lamp data sheet is not necessary, although it may be used in the measurement circuit as long as the input voltage can be adjusted to obtain the nominal current. Since the lamp current is controlled during this type of measurement, the value of the ballast impedance has minimal effect on the lamp's operation.

This method of measurement helps to eliminate several uncertainty factors that contribute significantly to reference measurements; for example, voltage measurement uncertainty, ballast impedance variations with respect to temperature, and the effect of the parasitic capacitance of the wiring and other parts of the system. Thus, results from fixed-current measurements can provide measurements of lower uncertainty.

Generally, performing both reference and fixed-current measurements on a given lamp should give similar results, as the lamp under reference conditions should operate at near the nominal current. However, sometimes the results of the two types of measurements can be significantly different—and thus, can be instructive for lamp and ballast design.

Annex B – Use of the “Peak” Method

To reduce the test time, it may be desirable to take measurements at peak light output. By careful correlation tests, it may be possible to relate peak to stabilized lumens by a constant, which will be unique to each lamp type. This correction factor will equal the *stabilized lumens* divided by the *peak lumens*. If lamps are read at peak light output and lumen data is presented in terms of the stabilized value, the peak measurement shall be multiplied by the correction factor.

The peak light output of some single-based fluorescent lamps may not be achieved at ambient temperatures of 25 °C (77 °F). Therefore, a heated photometric chamber will be required to provide a temperature higher than that at which light output is a maximum. This will ensure that a true peak reading is obtained as the lamp light output passes through it during the warm-up cycle. Generally, this temperature should not exceed 40 °C (104 °F).

If the lamp is being transferred from a stabilization position to the test position, the circuit switching procedure described in **Section 6.1.4** should be followed. The precautions in **Section 4.4** regarding airflow should be taken.

Since several lamp design and process variations exist, considerable testing and experience with a given lamp type may be required to correlate peak and stabilized measurement results. Therefore, the stabilized method is preferred. If the peak method is used, it shall be so noted in the test report.

Annex C – Normal Intensity (Candela) Measurements

The method described in this Annex was used previously when many laboratories did not have goniophotometers. This method does not require a goniophotometer and it is simple to set up and use.

Directional light output may be measured with one or more detectors at an appropriate distance from the lamp. The photometer detector shall have a relative spectral responsivity which approximates the $V(\lambda)$ function. The recommended minimum distance is five times the longest dimension of the lamp. Total luminous flux may be computed if the flux-to-intensity ratio for the lamp type is well established.

For detector sources of error such as spectral mismatch, the reader is referred to ANSI/IES LM-78-20.¹³ Other potential sources of error that require consideration are:

- Stray light
- Distance measurement
- Deviations from the inverse-square law (with respect to angular size of source)
- Angular alignment of lamp and detector

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